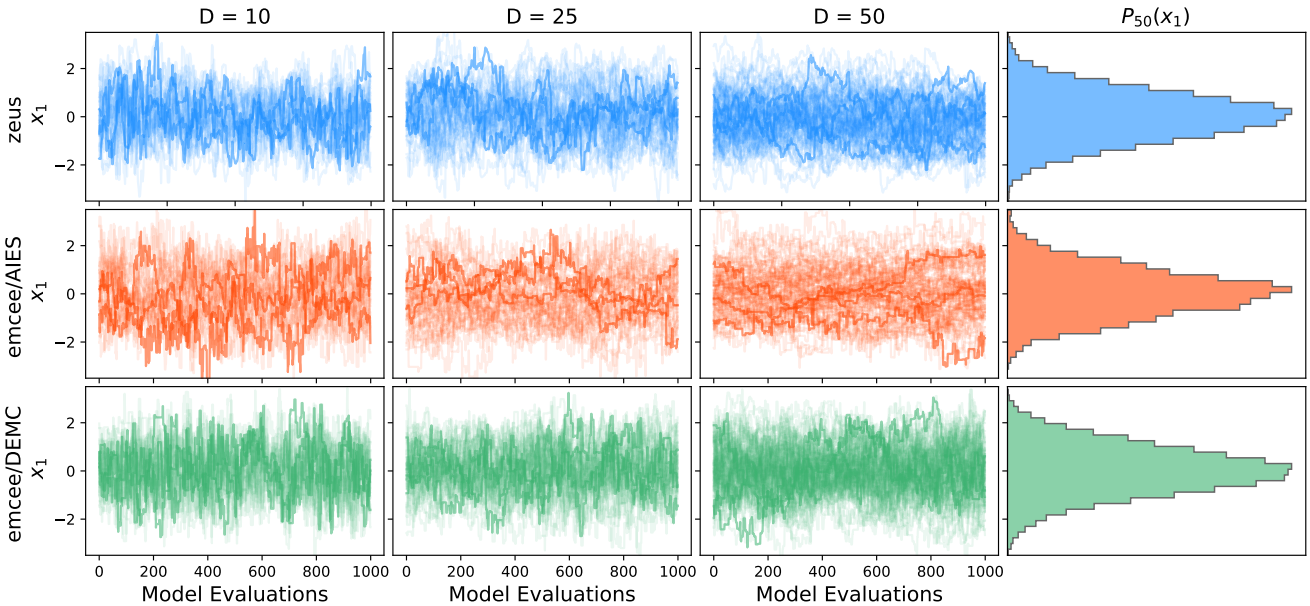

Figure 3. The figure shows numerical results (i.e. walker trajectories/chains for the first parameter) demonstrating the performance of the three ensemble MCMC methods in the case of a normal (Gaussian) target distribution in 10, 25 and 50 dimensions respectively. The last column illustrates the 1-D marginal posterior corresponding to the first parameter x_1 estimated directly from the samples for the 50-dimensional case.

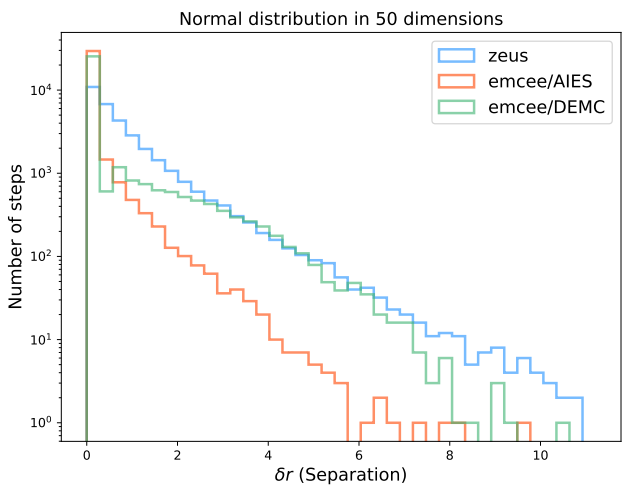

Figure 4. This figure shows the distribution of step sizes of walkers for the three different samplers in the case of a normal (Gaussian) target distribution in $D = 50$. It is important to note here that both `emcee` algorithms exhibit a peak at zero separation; `zeus` on the other hand does not due to its non-rejection nature.

distribution which by definition is characterised only by linear correlation between its parameters. The second toy problem is the ring distribution, a characteristic example of strong non-linear correlations. The third example is a Gaussian mixture with two components. While the purpose of the first two examples is to study the behaviour of the algorithm in the presence of linear and non-linear correlations respectively, the goal of the third example is to demonstrate the ability of `zeus` to sample efficiently from multimodal target distributions. The fourth toy example investigates the effect that heavy tails have on the sampling efficiency and the fifth the shows the effects that hard boundaries have on sampling.

We compare `zeus` with two popular alternatives offered by `emcee`, namely affine–invariant ensemble sampling with the *stretch* move (`emcee`/AIES) and the differential evolution move (`emcee`/DEMC). The main goal of this analysis is to justify our choice of slice sampling as the basis of `zeus` instead of Metropolis updates through the use of simple yet instructive toy examples.

For all three toy examples discussed below we adopt the same analysis procedure, where we initialise the walkers by sampling from a normal distribution $\mathcal{N}(\mathbf{0}, \mathbf{I})$ where $\mathbf{I}$ is the identity covariance matrix and we discarded 10^4 iterations as burn-in.

The main metric that we use to investigate the behaviour of the samplers in those toy examples and to compare their performance is the distribution of steps performed by the walkers. As a step we define the distance spanned in parameter space by a single walker in a single iteration. This is a fundamental measure of the efficiency of an MCMC method and it is directly related to the expected squared jump distance (ESJD; Pasarica & Gelman 2010) given by:

$$\text{ESJD} = \mathbf{E}\left[|\theta_{t+1} - \theta_t|^2\right] = 2\,(1 - \rho_1) \cdot \text{Var}_{(\pi)}(\theta_t)\,, \tag{2}$$

where θ_t are the chain samples, ρ_1 is the first-order autocorrelation, and $\text{Var}_{(\pi)}(\theta_t)$ is a function of the stationary distribution only. Assuming that the higher-order autocorrelations $\rho_2, \rho_3, \ldots$ are monotonically decreasing with respect to ρ_1, then maximising the ESJD leads to minimisation of the autocorrelation between chain elements and thus maximisation of the sampling efficiency. In other words, the further away (i.e. the greater the ESJD) the walkers jump per iteration, the higher the sampling efficiency of the method. A benefit of using ESJD instead of the autocorrelation time as a metric is that the former, as an expectation value, is more accurate when computed using short chains.

In order to account for the different computational cost (i.e. different number of model evaluations per iteration) between `zeus` and `emcee` we thinned the chains of the latter method according to the average number of model evaluations of `zeus`. This allowed us to compare the distribution of steps of the three samplers as shown in Figures 4, 9, 11, 13, and 15 for the five toy examples respectively.

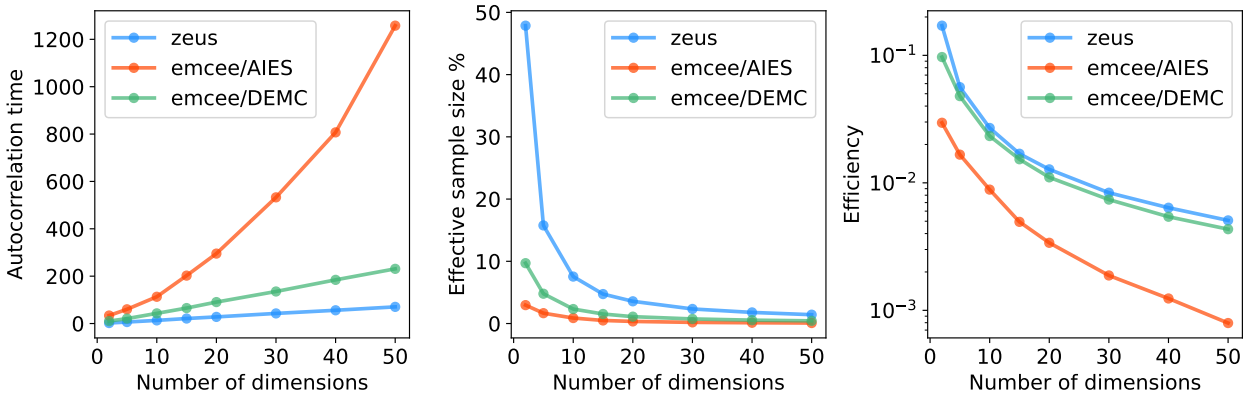

Figure 5. The figure shows numerical estimates of the integrated autocorrelation time (number of steps along a chain required to obtain an independent sample; left panel), the effective sample size (percentage of effectively independent samples in a chain; middle panel), and the sampling efficiency (i.e. effective sample size per model evaluation; right panel) for a normal target distribution and varying number of dimensions. The number of walkers was set to $4 \times D$ for `zeus` and $16 \times D$ for `emcee`, this was the optimal choice (i.e. the one maximising the efficiency for the given dimensionality) for each sampler. `zeus` and `emcee/DEMC` exhibit linear scaling of the autocorrelation time with the number of dimensions whereas `emcee/AIES` scales exponentially.

3.1.1 The correlated normal distribution

Starting with the normal target distribution it is important to note here that all three of the methods used in the comparison are affine–invariant[2], meaning that their performance is immune to any linear correlations between the parameters. Since the normal distribution incorporates, by construction, only linear correlations (i.e. the 2D marginal distribution contours look like ellipses), it is the perfect testing ground to assess the effect that high dimensionality has on the three methods independently of other complications. For our example we used a zero-mean normal distribution with a covariance matrix in which the diagonal elements are set to 1 and the off-diagonal ones are equal to 0.95. We then proceed by sampling the aforementioned distribution in 10, 25 and 50 dimensions. Based on Figure 3 one can see that the walkers of `emcee`/AIES dissolve into an inefficient random walk characterised by low step size and high autocorrelation time as the number of parameters increases. `zeus` and `emcee`/DEMC are not so severely affected by the high number of parameters exhibiting a substantially lower autocorrelation.

Let us now try to explain this difference in behaviour by looking into the distribution of the steps of the walkers in Figure 4. One thing to notice here is that the distribution of the steps of `zeus`'s walkers extends significantly further away than those of `emcee`/AIES and `emcee`/DEMC. This should come as no surprise since the construction of the approximate slice allows for larger steps than Metropolis updates as shown in Table 1. This is because when a proposal is rejected in slice sampling the approximate slice shrinks and another sample is proposed instead. In this way `zeus`'s walkers always move and the chance of staying fixed is zero – unlike MH-based updates in which frequent rejection of samples is a necessity. This aforementioned procedure leads to greater steps in parameter space. The difference between `emcee`/AIES and `emcee`/DEMC is attributed to the fact that DEMC uses a proposal scale[3] $\gamma = 2.38/\sqrt{D}$ that guarantees a constant acceptance rate accounting for the number of dimensions D. This proposal scale is however optimal only in the case of a normal target distribution such as the one that we are studying here and there is no guarantee that it would return acceptable results in non-Gaussian distributions. For the case of `emcee`/AIES, the relevant proposal scale γ is allowed to vary in the range between $1/\alpha$ and α where $\alpha = 2$ is often taken as the typical value. It is clear that in the latter case γ does not possess the desired scaling $\gamma \propto 1/\sqrt{D}$ and thus, although the method generates proposals in the right overall direction, most of the samples do not reside in the typical set (Speagle 2019). In other words, the lack of proper scaling of the proposal scale with the number of dimensions leads to `emcee`/AIES "overshooting" the typical set where most of the posterior mass is located.

Table 1. The table shows a comparison of `emcee`/AIES, `emcee`/DEMC and `zeus` in terms of the expected squared jump distance (ESJD; higher is better) for the five toy examples i.e. 50-D normal distribution, 25-D ring distribution, 25-D Gaussian mixture, 25-D Student's t-distribution, and 25-D truncated normal distribution.

	`emcee`/AIES	`emcee`/DEMC	**`zeus`**
Normal	0.5288	1.1162	**2.1354**
Ring	0.0043	0.0006	**0.1257**
Mixture	0.0037	0.0056	**0.1015**
Student	12.9124	2.4137	**23.5720**
Truncated	0.0940	0.3501	**0.5882**

We can also draw some useful insights about the sampling efficiency of those samplers and its scaling with the number of dimensions by estimating the integrated autocorrelation time of the chains. Given the autocorrelation time we can also estimate the effective sample size as the percentage of effectively independent samples in a chain. Dividing the effective sample size with the computational cost of each method we can then estimate the sampling efficiency. The results of such a comparison are shown in Figure 5. We immediately notice here that the autocorrelation times of `zeus` and `emcee`/DEMC scale linearly with the number of dimensions, whereas the autocorrelation time of `emcee`/AIES scales exponentially. The computational cost of `zeus` per iteration per walker, although somewhat higher than that of `emcee`, does not vary with the number of dimensions. This means that in high dimensions, `zeus` dominates over `emcee`/AIES in terms of sampling efficiency.

[2] Differential evolution Metropolis is only approximately affine–invariant due to the jitter that it is often added to its proposal. This however has a negligible effect.

[3] The proposal scale γ is similar to μ used in ESS in the sense that its value determines the length scale of the proposed jumps in parameter space. A high value would lead to large steps that are often rejected and a low value would lead to small steps that are often accepted but do not carry the walkers far. For such methods, a balance must me found.

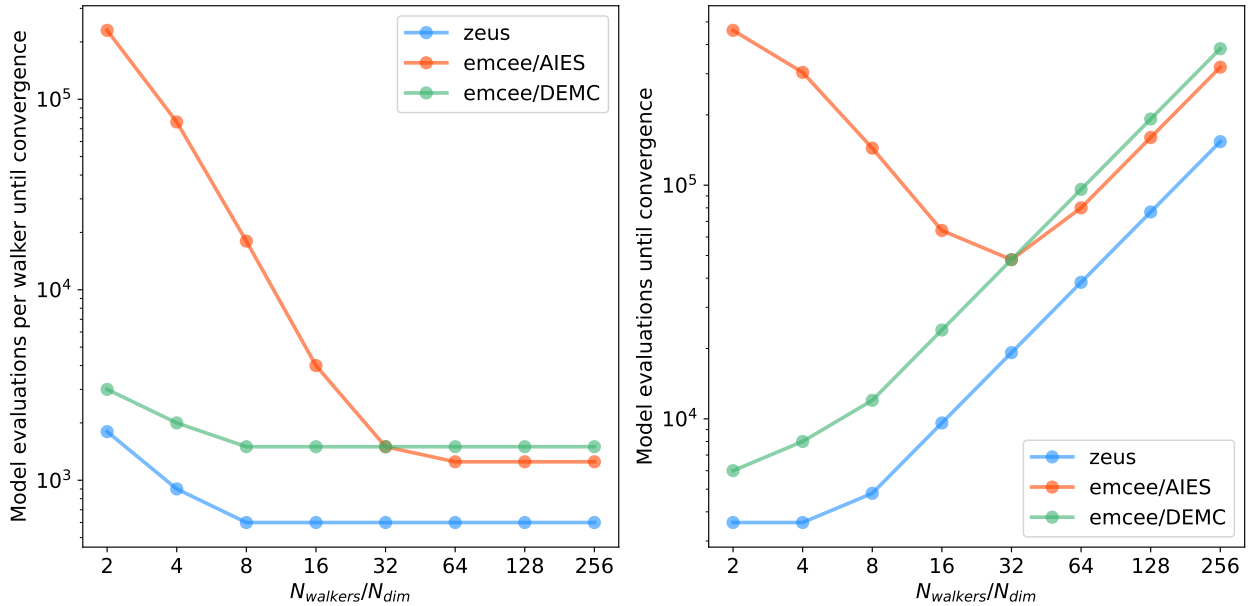

Figure 6. The figure shows the computational cost until convergence is reached in terms of the number of model evaluations for the different ensemble samplers for a highly correlated 25–dimensional normal distribution. The left panel shows the computational cost for a single walker. From this we can see that the cost for a single walker decreases as we increase the number of walkers until it reaches a plateau. The high computational cost for low numbers of walkers can be attributed to the low variety or sparsity of possible proposals; this is significantly higher for `emcee`/AIES. The right panel takes into account the linear scaling of the total computational cost as we increase the number of walkers and shows the total computational cost for the whole ensemble until it converges.

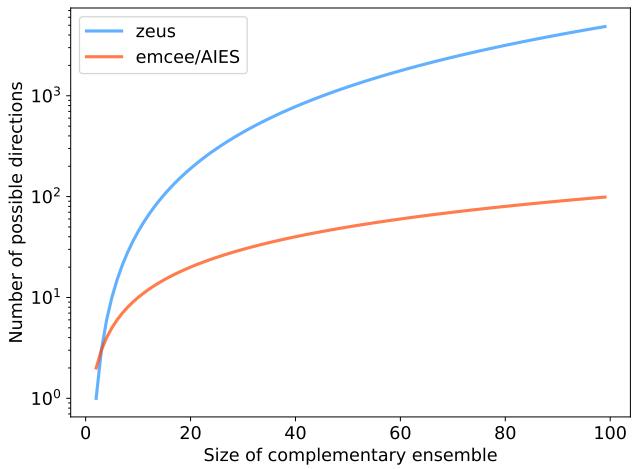

Figure 7. The figure shows the number of possible directions along which `zeus` and `emcee`/AIES can propose new samples as a function of the number of walkers in the complementary ensemble. `emcee`/DEMC exhibits the same number of proposals as `zeus` and it is not plotted here. `zeus` has a much higher variety of possible directions compared to `emcee`/AIES for any given number of walkers, assuming that that number is greater than 2.

The above discussion allows us to clearly state a crucial distinction between the three methods, which is their response to the curse of dimensionality. As the number of dimensions increases, the probability mass of a distribution is concentrated into a thin shell within the tails of the distribution (i.e. the typical set). To account for this and maintain its efficiency, a sampling method has to adjust its proposal scale – otherwise the proposals will not be located in the typical set and thus they will not be accepted. The three methods that we mentioned so far deal with this in different ways. `emcee`/AIES's proposal scale is not adjusted and thus its proposals become increasingly inefficient in high dimensions. `emcee`/DEMC's proposal scale is adjusted based on the theoretical expectation for the case of the normal target distribution. Although both `emcee` methods perform well in this example, their sub-optimal scaling will degrade their performance in non-Gaussian target distributions as we will demonstrate in the next toy example. Finally, `zeus`'s proposal scale is continuously adapted, as the slice expands and contracts in every iteration, thus guaranteeing optimal scaling. Huijser et al. (2017) found that the suboptimal scaling of `emcee`/AIES with the number of dimensions can introduce biases into the expectation values derived from the chains in high dimensions that are hard to diagnose. The locally adaptive nature of `zeus` allows it to avoid this problem by adjusting its proposals accordingly.

Another kind of analysis we can perform is to use the highly correlated 25–dimensional normal distribution as the target distribution and estimate the convergence rate of the three samplers. Although simple, the normal distribution is a valid approximation of many realistic astronomical posterior distributions and as such we expect the results presented in this paragraph to be applicable to a wide range of other distributions that resemble the normal distribution to some extent. We acknowledge however that the *no free lunch* theorem also applies to this case, and there are bound to be cases in which the results would be qualitatively different. That being said, we initialised the walkers from a compact normal distribution (i.e. standard deviation equal to 10^{-4} times that of the target distribution) centred around a point along the first axis of the parameter space at a distance of 100 standard deviations from the mode. We then measured the number of model evaluations required until the samplers have converged to the target distribution. The results for varying number of walkers are presented in Figure 6.

In general, walkers move along directions defined by the walkers of the complementary ensemble. Thus, increasing the number of walkers offers a wider variety of available directions along which the walkers of `zeus` or `emcee` can move via slice sampling or Metropolis updates respectively. This is demonstrated in the left panel of Figure